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New Methods and Reagents in Organic Synthesis. 14.¹⁾ A Simple Efficient Preparation of Methyl Esters with Trimethylsilyldiazomethane (TMSCHN₂) and Its Application to Gas Chromatographic Analysis of Fatty Acids

Trimethylsilyldiazomethane (TMSCHN₂), known as a stable and safe substitute for highly toxic and explosive diazomethane in the Arndt-Eistert synthesis and homologation of carbonyl compounds, has smoothly reacted with various carboxylic acids in methanolic benzene solution to give the corresponding methyl esters in excellent yields.

Keywords—trimethylsilyldiazomethane; carboxylic acid; methyl ester; esterification; fatty acid; gas chromatographic analysis

Preparation of methyl esters of carboxylic acids, especially for analytical purposes, has been generally carried out²⁾ by the reaction of carboxylic acids with (1) methanol in the presence of acidic catalysts such as sulfuric acid, hydrogen chloride, boron trifluoride *etc.* which are mostly corrosive, (2) diazomethane which presents an explosive hazard in addition to its high toxicity, or (3) dimethylformamide dimethylacetal or trimethylphenylammonium hydroxide which requires heating for esterification.

We have reported recently that trimethylsilyldiazomethane (TMSCHN₂),³⁾ which is a stable and safe substitute for hazardous diazomethane, can be efficiently used for the Arndt-Eistert synthesis⁴⁾ and homologation of ketones⁵⁾ and aldehydes.⁶⁾ We now wish to report that the reaction of TMSCHN₂ with carboxylic acids in the presence of methanol quickly gives methyl esters in excellent yields at room temperature and the method can be efficiently applied to analytical works such as determination of carboxylic acids by gas chromatography:

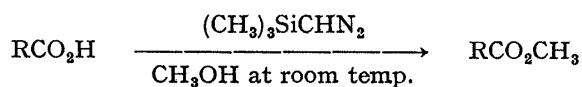


TABLE I. Esterification of Carboxylic Acids with TMSCHN₂ and Methanol

Run	RCO ₂ H	Isolated yield of RCO ₂ CH ₃ , %
1	2-Nitrobenzoic acid	95
2	Piperonylic acid	Quant.
3	Salicylic acid	Quant.
4	Mesitoic acid	97
5	Picolinic acid	82.5
6	Thiophene-2-carboxylic acid	Quant.
7	Cyclohexanecarboxylic acid	Quant.
8	Palmitic acid	Quant.
9	Oleic acid	Quant.
10	α-Ketoglutaric acid	85
11	Cholic acid	Quant.
12	L-Phenylalanine ^{a)}	42 ^{b)}

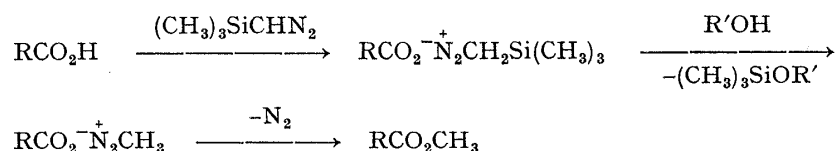
a) Reaction time was 4 hr.

b) As the hydrochloride. The starting material was recovered in 22% yield.

A general experimental procedure for the new preparation of methyl esters is as follows: To a stirred carboxylic acid (1 mm) in methanol (2 ml)–benzene (7 ml) was added TMSCHN₂⁷⁾ (1.3 mm) in benzene (1 ml) at room temperature. The mixture was stirred for 30 min at room temperature and concentrated to give the corresponding methyl esters.

The esterification proceeds instantaneously and quantitatively when an excess of TMSCHN₂ is used, and the reaction can be easily monitored by the disappearance of the yellow color of TMSCHN₂. The utility of the procedure is well demonstrated in Table I. Various carboxylic acids including aromatic, heteroaromatic, alicyclic, and aliphatic ones smoothly undergo the esterification. Salicylic acid and α-ketoglutaric acid easily give the corresponding methyl esters without any change of their phenolic and keto functions, respectively. The basic nitrogen of picolinic acid as well as the double bond of oleic acid is completely intact to TMSCHN₂. Cholic acid also undergoes the esterification with the retention of its hydroxylic function. L-Phenylalanine affords the corresponding methyl esters accompanied with some of the starting amino acid.

Although 2-nitrobenzoic acid reacted with TMSCHN₂ in 20% methanolic benzene to give methyl 2-nitrobenzoate in 95% yield, replacement of methanol with ethanol resulted in a little decrease of the yield (87%) of the methyl ester and a small amount (5%) of the trimethylsilylmethyl ester⁸⁾ was obtained. The latter ester was the main product (86%) when 20% *tert*-butyl alcohol–benzene was used as a reaction solvent. Furthermore, TMSCHN₂ does not give any diazomethane when it is treated with 20% ethanolic benzene. These results deny the participation of diazomethane which might be formed from TMSCHN₂ by the action of methanol⁶⁾ and suggest the reaction mechanisms⁸⁾ as follows:



The method is applicable to gas chromatographic analysis of fatty acids.⁹⁾ Thus, a mixture of six fatty acids (lauric, myristic, palmitic, stearic, arachidic, and behenic acids) is quickly and quantitatively esterified with TMSCHN₂ in 20% methanolic benzene at room temperature. The esterification mixture, after adjustment to the desired concentration or solvent, is ready for gas chromatography. Its standard calibration curves relating peak area ratios (each fatty acid/palmitic acid as internal standard) to mole concentrations hold

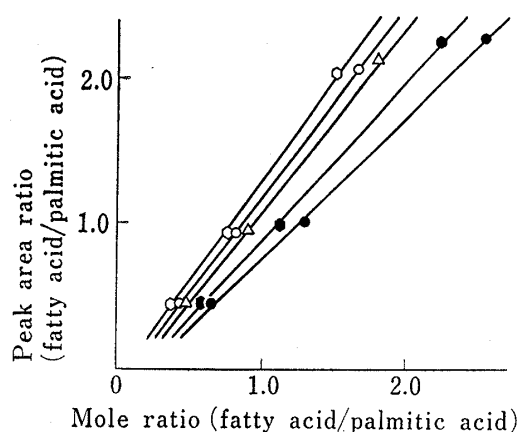


Fig. 1. Standard Calibration Curves for Determination of Fatty Acids

(●): lauric acid, (●): myristic acid,
(△): stearic acid, (○): arachidic acid,
(○): behenic acid.

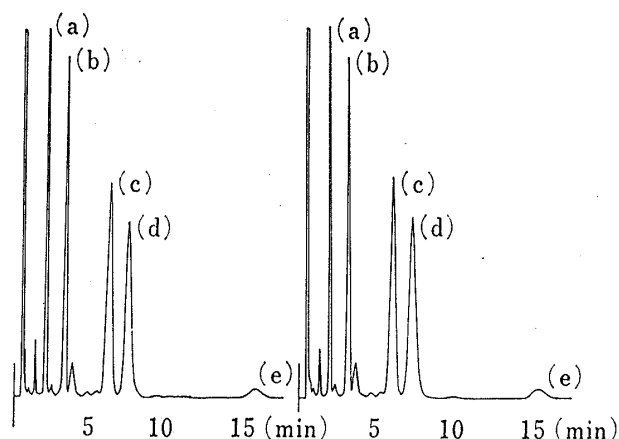


Fig. 2. Gas Chromatograms of Fatty Acid Methyl Esters obtained by the TMSCHN_2 (left) and Diazomethane (right) Methods

(15% EGSS-X, $1\text{m} \times 5\text{mm}\phi$ glass column temp., 200° ;
 N_2 ; FID; chart speed, 10 mm/min).
(a): myristic acid, (b): palmitic acid, (c): oleic acid,
(d): linoleic acid, (e): arachidonic acid.

the linear relationship in a wide range (Fig. 1). Furthermore, the gas chromatogram of methyl esters prepared from a mixture of five fatty acids (myristic, palmitic, oleic, linoleic, and arachidonic acids) by the action of TMSCHN_2 and methanol is completely identical with that of methyl esters prepared by the action of diazomethane (Fig. 2). These facts prove an identical potency of TMSCHN_2 with diazomethane in the gas chromatographic analysis of fatty acids and promise a non-hazardous routine procedure for the preparation of methyl esters of fatty acids in small quantities.

Esterification of carboxylic acids with TMSCHN_2 -methanol is conveniently and safely carried out with superior efficiency. Since the reaction quickly proceeds under very mild conditions (room temperature and essentially neutral medium), the method should have a broad applicability not only to preparative but also for analytical works.

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Quantitative Analysis of Clebopride and Its Metabolites in Rat Blood by Acid Decomposition

6-Chloro-*m*-anisidine was obtained as one of the products by the acidic decomposition of elebopride and its main metabolites (total CP). This degraded product after derivatized with heptafluorobutyric anhydride was measured by use of mass fragmentography. A good calibration curve was obtained in the range 2–80 ng/ml of clebopride in rat blood. The concentration of total CP in rat blood after intravenous (0.1 and 0.5 mg/kg) and oral (0.5 mg/kg) administration was determined by this method.

Keywords—clebopride; mass fragmentography; GC/MS; acid decomposition; benzamide drug; pharmacokinetics

Clebopride (CP; N-(1'-benzyl-4'-piperidyl)-2-methoxy-4-amino-5-chlorobenzamide) is a new benzamide drug with potent antidopaminergic activity.¹⁾ Therapeutic dose of CP is low (0.5 mg/kg). This drug is extensively metabolized as shown in Table I.²⁾

Segura *et al.* recently reported an analytical procedure for CP in biological fluids³⁾; however this procedure meet only to need for determination of the plasma level following doses more than 10 mg/kg in the experimental animals.

The present report describes mass fragmentographic determination of the unchanged CP and its main metabolites (total CP) in whole blood. This method is based on the fact that not only the unchanged CP but also its main metabolites having 6-chloro-*m*-anisidine (6-CA) ring have to be decomposed with hydrochloric acid to form 6-CA as one of the degraded

TABLE I. Structures of Clebopride, Its Main Metabolites and Internal Standard

	R ₁	R ₂
Clebopride		CH ₃
Metabolite 1		CH ₃
Metabolite 2		CH ₃
Metabolite 3		CH ₃
Internal standard		CH ₂ CH ₃